Institutions from Interaction: The Emergence and Transmission of a Division of Labor in Coevolutionary Societies

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Abstract

Institutions, in Weber's sense, are stable patterns of social action sustained by shared expectations. This paper grows them. In societies of autonomous learning agents that coordinate through emergent signs under a designed market mechanism, a division of labor of eleven to twenty professions arises in every converged society, together with supply chains, recorded employment, learned prices, and displayed signs that carry what their bearers deliver. The theoretical engine is symbolic interactionism, and the market is its economic instance. The construction is offered as a member of a new family of artificial intelligence methods, social algorithms, whose learned object is a social structure rather than a model. Agents of the evolutionary substrate begin with random genomes and no pretrained knowledge, so the institutions that appear can only be products of the interaction. On the language-model substrate, every channel except sign-mediated trade is closed by construction: market societies outproduced a single long-context model on identical demands, and where capability was hidden from every participant, the market's reward ordering recovered the hidden ranking while a blinded coordinator recovered nothing. A reconstruction panel shows that a society's division of labor is transmissible through its measured statistics alone: seeded into fresh societies, a source's statistics regrow its professions, so that treatments can be rehearsed on a replica before the world.

1 Introduction

This paper practices computational social science in the constructive mode. If institutions emerge from symbolic interaction (Mead 1934), then building the interaction should grow them, and what grows can be measured, identified, and transmitted. We build societies of autonomous learning agents under a designed market mechanism and report what arises, Weber's institutions, read with the instruments of the field.

The construction also belongs to a family. Every established approach to artificial intelligence copies a natural system that learns: genetic algorithms copy evolution, neural networks copy the brain, and reinforcement learning copies conditioning. This paper develops a fourth member, *social algorithms*, artificial intelligence that copies society's own meaning-making, here in its economic instance, the market. Human markets are a learning system in their own right: who should make what, which skills deserve deepening, and what everything is worth, learned by thousands of specialists coordinating through nothing but prices and displayed signs. We build that system artificially and let the market do the work other paradigms assign to a loss function or a reward signal: credit assignment, exploration pressure, the shaping of what is learned.

The copy is close, because the rules of the designed market implement laws of social science. Smith's law, that the division of labor is limited by the extent of the market (Smith 1776), becomes a dense demand field, without which no division has room to form. Ricardo's comparative advantage becomes every component being purchasable, and the recorded employment below is that choice happening. Hayek's observation that prices aggregate dispersed knowledge (Hayek 1945) is enforced by construction: prices and signs are the only shared channels. Spence's condition, that signals stay informative about unobservable quality only while the unqualified cannot afford to sustain them (Spence 1973), becomes the rule that worthless work earns nothing. Smith's invisible hand receives its mechanism: it *is* the emergent shared meaning, a sign system annealed across generations by the utilities of all the ancestors. Throughout the paper, an *institution* is a stable pattern of social action sustained by shared expectations (Weber 1978), and the market mechanism itself becomes one in Berger and Luckmann's sense (1966), externalized in signs and offers, objectivated into social facts, internalized in every next act.

What the paradigm learns is a social structure rather than a model: professions, supply chains, and prices carrying information no single agent holds. It is also learner-agnostic: the market mechanism is the method, and the learner inside each agent is pluggable, an evolution strategy on one substrate and a large language model on the other. Coevolution is the operative principle, since each agent's environment is mostly other agents and every improvement reprices opportunity everywhere. Beneath both runs the paradigm's hermeneutic principle, the *autonomy of perception*: every agent interprets signs by its own experience alone, and no meaning is given, copied, or shared except through interaction itself.

The design contributes an identifiability strategy for emergence. Evolutionary agents begin with random genomes, no pretrained knowledge, and meaningless signs, so when a division of labor, employment, learned prices, and sign conventions appear, there is nowhere else they could have come from. Language-model agents arrive saturated with human social knowledge, so for them the blinding is the identification instead: capability invisible to everyone, individual recognition banned, rosters anonymous, every channel but sign-mediated trade closed by construction. When the market's reward ordering recovers the hidden capability types (Section 10), the market itself must be what found them, because every other route was closed, and running the same mechanism on both substrates separates what the mechanism produces from what the agents brought.

The design closes the micro–macro loop in both directions. Upward, autonomous interpretation generates the macro level, none of it encoded, which is emergence in Coleman's sense. Downward, the emergent institutions govern individuals: prices reprice every opportunity, rising aspirations discipline delivery, and reward ordering assigns individual fates. Reconstruction (Section 11) runs the loop as an experiment. The paradigm is evaluated against its neighbors, evolution against coevolution and societies against single minds, with every control the strongest standard alternative and all else held identical, and every measurement below is defined by its computation before any claim rests on it.

2 The world

2.1 What a society is

A society is one running simulation with two kinds of participant. *Learners* construct products from a catalog, sell them, and buy: a learner's product may incorporate a component purchased from another learner, with payment cascading down the chain when the final product sells, which makes supply chains and employment possible. *Buyers* are the demand field, standing in for the humans the society serves: each holds one want, an authored budget expressing how strongly the society holds that need, a Cobb-Douglas utility over quality received and budget kept, and an offer price that is learned, never assigned. Trades settle by a band double auction: each side offers a learned band, a midpoint plus or minus a range; no intersection, no sale; otherwise the price is the midpoint of the intersection, scaled by quality as defined below.

2.2 The catalog (evolutionary substrate only)

Learners assemble programs from a catalog of real machine-learning tools organized as a tree, with the parameter settings of high-leverage tools walkable as further levels of the same tree, one hundred ninety-one production leaves in all. A plan that stops its walk at a tool expresses it at library defaults, so partial expertise is expressible and settings can become professions; full mechanics are in the ODD document.

2.3 Wants, tests, and grounded scores

Every buyer's want has three parts: the item sought (say, a clustering program), one or more measuring instruments, and a price level. The instruments are ordinary machine-learning evaluations; behind every classifier claim, for example, is the F1 score on held-out data, where the delivered program must label a testing portion it never saw and F1 combines precision and recall into one number between 0 and 1.

When a buyer examines a delivery, the chain is actually executed: the programs really run, on real corpora, and failures really fail. A chain whose evaluation crashes, or whose score is zero, earns nothing. There are no consolation payments anywhere in these societies: if worthless work earns even a little, agents learn to display the outward signals of quality without the quality, and the signaling system rots.

2.4 Signs: how agents find each other

Every agent displays one sign, a vector of eight floating-point numbers, a position in meaning space; every trade plan carries a sought sign of the same form, the position it searches, and signs and offers post to one shared board. Matching is by cosine similarity in that space, the buyer's sought sign against the seller's displayed sign and the seller's sought against the buyer's displayed, blended half and half. No meaning is assigned anywhere: whatever a sign comes to mean, it means because of how its bearer traded.

2.5 Attention is scarce

A buyer does not examine every offer. Offers are audited in sign-similarity-weighted random order, and auditing stops when a delivery passes; on the language-model substrate the audit budget is five per want per period. Individual recognition, memory attached to an identity rather than to a sign, is banned by design. A product can therefore fail because it is bad or because it never wins an audit.

2.6 Money

Payment is quality-scaled. The best score a buyer has verified for its want defines that want's *frontier*; the frontier earns the agreed price, and weaker work earns the fraction given by its score over the frontier. Learners buying components from learners creates payment cascades down chains.

Every society reported here runs with live payment transfer and a full ledger of time, payer, payee, and amount; every employment claim below rests on those receipts.

2.7 The two substrates

The *evolutionary* substrate gives each learner a covariance matrix adaptation evolution strategy over its trade plans, with market outcomes as fitness. The *language-model* substrate replaces it with an agent that reads its market situation each period and writes code deliberately. Language-model agents are individually far more capable, which is exactly why they are informative: they isolate what comes from the market mechanism rather than from the difficulty of learning to program.

Reproduction kit: the engine, every society configuration, the language-model harness, the instruments, the hidden capability casts, the frozen reconstruction references, and the results tables are public at `https://github.com/agentbasedlearningsystems/institutions-from-interaction`.

3 The designed rules of a market that learns

The mechanism carries many interacting rules; four help most visibly in what follows.

A large catalog. One hundred ninety-one production leaves, with parameter settings walkable as leaves of the same tree, so that distinct trades are expressible at all. A catalog is the space of expressible skills; without room to differ, no division of labor can form.

Dense demand. Forty-two buyers spanning embedding, clustering, anomaly detection, classification, and retrieval work across eleven corpora. The wants form a ladder: entry rungs priced within reach of a newborn agent's first random success, upper rungs judged by several instruments at once or across two corpora. The field gives nearly every viable early behavior its own first customer; early differentiation survives.

Nothing for worthless work. A delivery scoring zero is refused outright: no sale, no consolation. An agent paid despite failing learns to display the signs associated with success, so the signs stop meaning success, and buyers lose the ability to find quality at all.

Composition wanted above its parts. No price is authored anywhere; what is authored is utility. A want whose product takes several stages to make carries a bigger budget than the wants for its parts, so a higher price for composed work can be earned in the market rather than set by the experimenter.

4 Result: a division of labor, and what it consists of

A trade here is a learner's primary product, its most frequent settled delivery, and evenness is normalized entropy over practitioners per trade. The census is taken at convergence: each society's last hundred settled deliveries. Across the seventeen societies whose converged window is full, every society has a division of labor: eleven to twenty distinct trades per society at evenness 0.91 to 0.98 (Figure 1), and at its widest read the eldest practiced twenty distinct trades over twenty-three learners at evenness 0.98; the rest had not reached a hundred deliveries. The division includes settings as professions: distinct settings of the same tool are sold as distinct trades, and walked-settings deliveries appear in every society (division census in the reproduction kit). The solitary controls, one learner under the identical demand field and catalog, ask whether a society reaches software quality a single learner cannot; the comparison sits with the improvement results in Section 9.

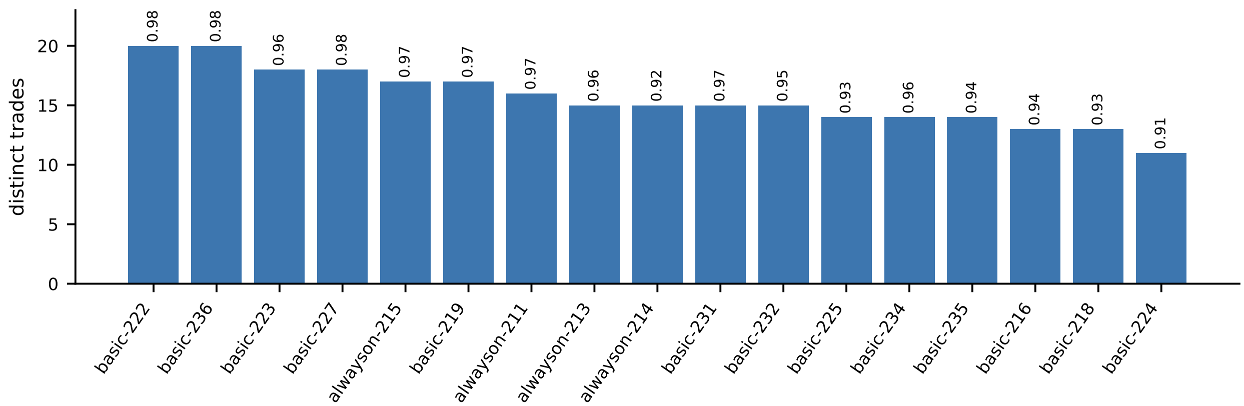

Figure 1: Distinct trades per censusable society, with evenness printed above each bar; per-society values are in the reproduction kit.

5 Result: prices are learned at both ends

Every buyer's offer band is learned with its Cobb-Douglas utility as fitness; authored budgets express how strongly the society holds each need. Buyers learn thrift where work is cheap to replace: entry-level offer midpoints fall in twenty of twenty-eight societies. Composed work commands more than entry work in the median society, and that gap is widening in nineteen of twenty-eight (sign test $p = 0.044$): the market is learning to pay more for what takes more.

6 Result: signs stay informative, and settle into local conventions

Informativeness is measured as the mutual information between a seller's displayed sign, its one public face, quantized to eight clusters within each society's recent window, and the family of product it lists: high mutual information means the sign narrows what its bearer offers. Across the twenty-nine societies measured, this reads 0.69 to 1.05 bits (median 0.74). Payment tracking quality keeps displayed signs attached to delivered reality.

Meaning also attaches to families of algorithms: in twelve of twenty-seven testable societies, *different sellers* of two settings-variants of the same algorithm display measurably more similar signs than different sellers of unrelated same-role algorithms (pooled listings over recent boards, permutation over algorithm-family labels, same-seller pairs excluded, each $p < 0.05$). Filing settings variants near one another is a recurring local convention.

7 Result: employment

The ledger answers the employment question with receipts rather than inference: does one learner ever *employ* another, paying it for a component used in a product it then sells?

Yes. In the most recent quarter of trading, learner-to-learner payments appear in fifteen of the twenty-seven ledgers measured (Figure 2). The share is small and heterogeneous, zero to 19.8 percent of end-window payment value, and higher in the end window than over the lifetime in ten of twenty-seven. Employment, where it takes hold, arrives late and grows; the strongest society already pays a fifth of end-window value through it.

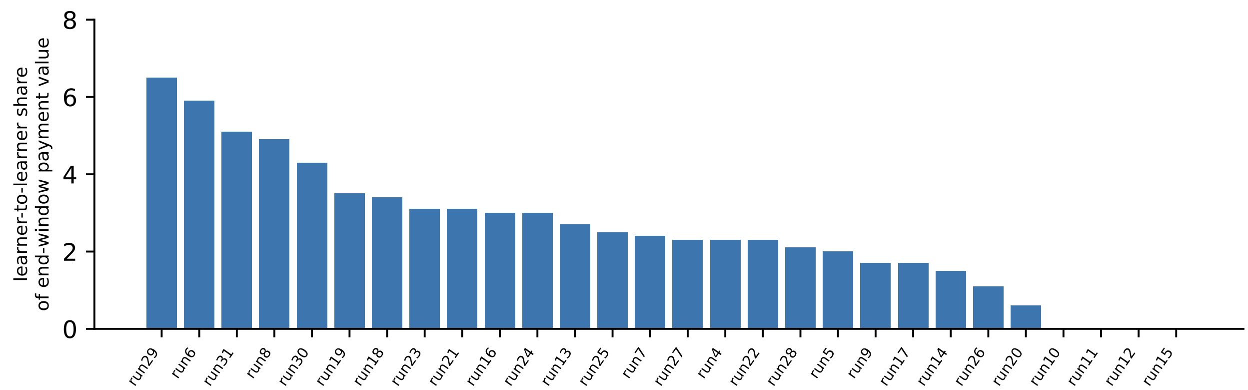

Figure 2: Learner-to-learner share of end-window payment value across the twenty-seven standard societies measured; fifteen show it, rising late in ten.

8 Result: hard demand stands open until it is earned

Demand for things the society cannot yet make is posted anyway, from the first iteration, at full price. Nothing schedules it and nothing needs to. A want whose product requires mastered components goes unfilled, leaking no money, until the society has learned enough to fill it: the frontier recedes exactly as fast as it is earned, and the market discovers the order of learning by itself.

The evidence is each want's first-settlement iteration. Entry-level wants settle early: median first settlements sit within a few dozen iterations in most societies, 115 at the widest, against lifetimes up to two thousand, while the solitary controls reach their first settlements only near iteration nine hundred. The upper rungs remain standing open in nearly every society after hundreds of iterations. The ordering, entry before upper and the hard rungs last or never, holds in twenty-eight of thirty-one societies; the three exceptions are the youngest, too young to have settled anything. None of it was scheduled: every want stood priced from the first moment.

9 Improvement runs to the ceiling of the scenario

Measured naively, improvement can look stalled: best-ever scores are set early and average sale quality drifts. Decomposed by tier, entry against the rungs above, the drift is composition. At convergence, in each society's last hundred deliveries, near-worthless deliveries sit only at the cheap entry wants, a median 6.5 percent of entry-tier settlements across the twenty-five societies measured, while the tiers above stay clean, at median zero. The frontiers sit at the practical ceilings of their tasks, and the rungs beyond are still climbing. Improvement runs to the ceiling of the scenario; unbounded improvement requires demand always one mastery beyond the society's reach, which standing unfilled wants provide by construction (decomposition in the reproduction kit).

The solitary controls return a mixed verdict. At equal society time the group dominates: by round four hundred, both solitary learners sit below all eleven comparable societies (exact $p = 1/78$). At equal total learning effort, the lone optimizer reaches comparable quality (0.17 and 0.29 against 0.20–0.34). The group's advantage on this substrate is time, not quality per unit of learning: parallel specialists serve the same demand field far sooner.

10 The language-model societies

Every mechanism above also ran on the second substrate: twelve language-model agents per society, forty trading periods, the same corpora and instruments, scarce audited attention, quality-scaled payment with a rising expectation bar (the buyer's aspiration, a moving average of recent accepted quality), and component payments. Four societies ran under this mechanism.

The substrate's controls test the organizing mechanism itself. On the same demand sheet, a single long-context mind given three hundred periods reached a mean best score per want of 0.377, serving four of nine wants. The market societies reached 0.460 to 0.758, serving up to eight of nine, all four above the single mind, the best at twice its score and twice its coverage.

One structure appeared in every society run: the spontaneous *wholesale supplier*. In four of four societies, some agent specialized in supplying one humble upstream component, a document embedding, to the others' retail products, and in all four that wholesaler ended as its society's top net earner. The strongest sold its component 194 times to ten distinct consumers, 51 percent of its society's component market; across the four, shares ran 39 to 55 percent with nine to eleven consumers. The role replicates out of sample: under a criterion fixed on these four societies, six of nine mixed-capability market societies pass it with two near misses, the cross-corpus rung is served in all nine, and the top component supplier is also the top net earner in seven of nine. The niche is open rather than reserved, held by large and small hidden models about equally; it exists because attention is scarce and components are purchasable, comparative advantage priced into existence rather than prompted.

Two further results follow, one positive and one instructively negative. The generalization rung, one delivered classifier scored on two corpora and paid the worse of the two scores, was served in all four societies, and the quality of those deliveries climbed to about 0.75 of a possible 1.0 in each: cross-domain generalization, reproduced four times. The deduplication rung, an end task rewarding almost pure data cleaning, scored by recovery of disguised duplicate rows planted against a hidden key, sold zero times in all four, and the agents' own planning text explains the refusal as economics rather than inability: a rung unfamiliar to every agent, priced like familiar ones, was rationally deferred all run; unfamiliar trades need premium prices or longer horizons.

A further experiment asks whether a decentralized sign economy can find hidden talent no allocator can see. Each society mixes eleven agents from three capability tiers (five small, four middle, two large), the assignment hidden from every agent, coordinator, and market mechanism; displayed signs are the only stable public face, and individual recognition is banned in both forms; this free-sign condition is arm A. The same hidden cast lives twice: under the market mechanism, and under a coordinator assigning work from a freshly shuffled anonymous roster of signs and latest deliveries, a coordinator that sees exactly what market participants see: displayed signs and delivered work, nothing more. In three of three market societies a large-capability agent finished as top net earner, and capable agents held mean within-society income ranks of 3.5 to 4.3 against 8.0 to 9.0 for the small tier, a pooled gap of 4.9 ranks (permutation $p < 10^{-4}$, thirty-three agents). Under the anonymous coordinator the same casts did not sort at all: rank gap 0.12 ($p = 0.47$), and in one society a small-tier agent was the top producer. Signs plus prices found the hidden talent every time; assignment without recognition found nothing (Figure 3). Runs stopped at a fixed twenty-period checkpoint.

The staged-sign arms return to the two labor-market questions of the 1991 program in Related work: is costly display sustained only where signs carry economic returns, and does a birth marker with no real difference behind it come to organize reward? In arm B, two sign dimensions display only at a cost, paid from market wealth or, in the team form, from wages: the suit that only earnings sustain. Market workers buy the costly dimensions and hierarchy workers never do, eighteen of

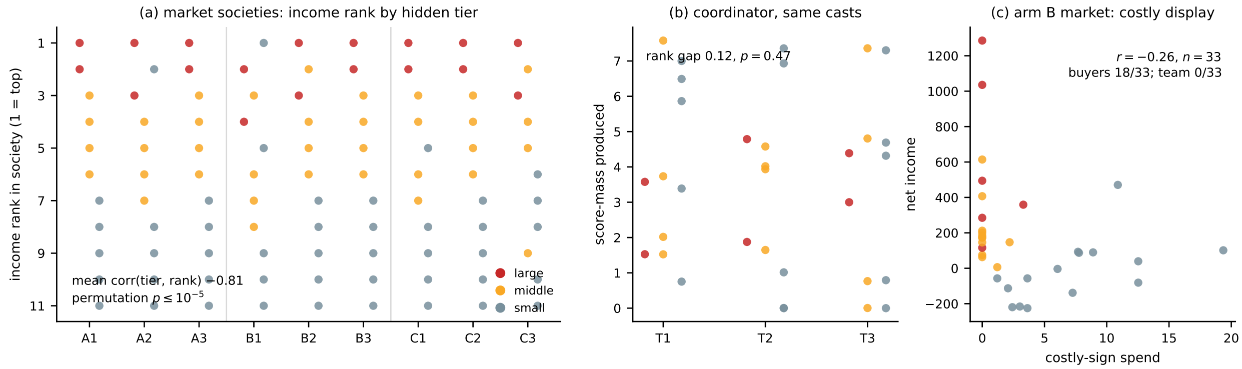

Figure 3: Hidden capability and reward across mixed-cast societies: (a) income rank by hidden tier, nine market societies, arm A the first three columns; (b) the arm A casts under the anonymous coordinator, score mass per worker; (c) arm B market workers, costly-sign spend against net income.

thirty-three against zero of thirty-three (Fisher exact $p = 1.5 \times 10^{-7}$; Figure 3c), with spending showing a weakly negative earnings association ($r = -0.26$). Costly display is bought where signs carry economic returns, and by strivers more than champions. In arm C, one sign dimension is fixed at birth, with capability tiers split evenly between markers, so the marker carries no actual difference between groups. The within-tier reward contrast between markers is null in both forms (market 0.65 ranks, $p = 0.27$; hierarchy 0.10, $p = 0.88$): at this horizon, against a test powered only for large effects, no differentiation developed around the meaningless marker. Across all nine market societies the sorting generalizes (Figure 3a): the mean within-society correlation between hidden tier and income rank is -0.81 (permutation $p \leq 10^{-5}$), and the top earner is a hidden large-capability agent in eight of nine, the ninth being the small-tier wholesaler.

11 Coevolutionary reconstruction: is a social state transmissible?

One further question drove the platform's design. Given the *measured statistics* of one society, and nothing else, no code, no agents, no private knowledge, can a freshly initialized society be grown into *that society's* social state rather than some other? In coevolutionary reconstruction, statistics from a source society seed a new society's starting dispositions, the market mechanism produces whatever it produces, and fidelity is judged by which source the daughter's mature state resembles. The question needs distinct social states to tell apart, and the standard societies provide them: the two sources chosen for the panel share only two trades, at total-variation distance 0.913.

The procedure, the data absorption technique (Duong 2013), takes statistics measured from a source and regrows them in a fresh simulation. Here a simulated society stands in for the real world as the source, so the reproduction itself can be tested: if a source's statistics regrow its social state, the same procedure can be pointed at statistics measured from a real population, and a treatment can be tried on the regrown society before it is tried on the world. It is also where the field's two traditions meet: theory-driven simulation supplies the generative micro level, data supplies the macro statistics, and the *diagnostic gap*, the difference between the behaviors a daughter was seeded with and the behaviors it sustains on its own, scores the theory against the data. Seeded behaviors the daughter keeps practicing are behaviors the motivational model can regenerate; seeded behaviors whose frequencies fall away are behaviors it cannot. Shrinking that gap by enriching the micro level is how neural architectures were built against benchmarks.

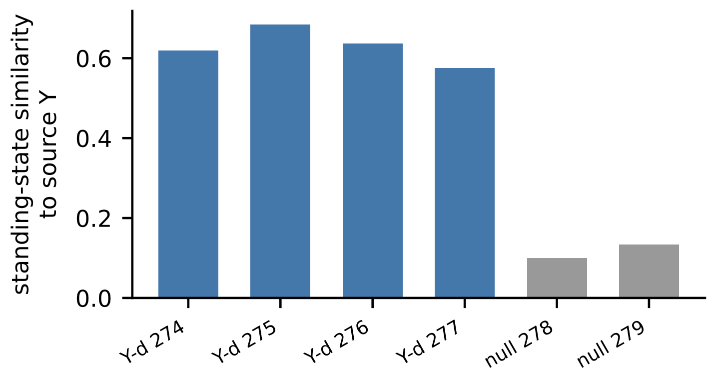

Figure 4: Reconstruction first reads: standing-state similarity to the seeding source, four qualifying daughters (blue) against two unseeded controls (grey).

From each source we census its *standing state*, every tool in the paid deliveries of its final five iterations, and write it into each newborn daughter's catalog as conditional probabilities at every tree level: ninety percent follows the census, ten percent stays uniformly random, no tool ever closed off. Nothing else transfers: no genomes, no prices, no signs, no memories. Four daughters are seeded from each source, and unseeded controls run the untouched catalog. The read, fixed before any daughter data existed, censuses each daughter's standing state against both frozen sources; success is every daughter nearer its own source (exact one-sided $p = 1/70$ at four per source), and controls define background drift.

Figure 4 holds the first panel's reads. All four daughters seeded from the dispersed source and past the read age reproduce their source's standing state at similarity 0.575 to 0.684, while the two unseeded controls grown to the same age resemble it at only 0.100 and 0.133, and the other source at zero: drift accounts for at most a fifth of what the seeded daughters show, and the rest is transmission. A confirmation wave followed, five fresh daughters from the same frozen census and five fresh controls, read once at age 300. All five daughters qualified and read 0.58 to 0.66; the four controls that reached the age read 0.00 to 0.17, no overlap (exact rank-sum $p = 1/126$). On this evidence, a division of labor is transmissible through its measured statistics alone. The concentrated source's daughters iterate more slowly, their seed pointing at costlier tools, so the complete two-source panel reaches its registered read only when every daughter has passed the read age; its result is not claimed here.

12 Limitations

The sample comprises twenty-eight standard societies and three solitary controls on the evolutionary substrate, and twenty-two language-model societies across the full-mechanism, mixed-capability, and staged-sign programs. The evolutionary societies span a wide age range; every number is a snapshot of a growing cohort. The division census requires a full converged window; eleven standard societies met that bar at submission, the youngest had settled nothing, and the census can only widen as the cohort matures. The reconstruction reads and the pricing read were fixed in advance: the reconstruction censuses and read ages were set before any daughter data existed; the pricing figures come from a single read declared the day before it ran, with earlier looks that day logged and none replacing it. The one pricing prediction fixed in advance, a positive composed premium in a threshold number of societies, was not met at that read and is not claimed; Section 5's widening gap is that read's secondary trend measure.

13 Related work

This platform continues a symbolic interactionist simulation program begun in 1991. The SDSS (Duong and Reilly 1995) modeled status symbols, prejudice, and social class with employer and employee agents reading and displaying three kinds of sign (unchangeable characteristics, signs bought with money from past employment, and freely changeable signs), with talent hidden at the interview. Its central results were that sign meanings converge across readers even when incorrect, that an arbitrary marker can come to mean incapacity, a vicious circle that is the emergence of social class, and that costly signals only the successful sustain become status symbols. A 1996 coevolutionary economy (Duong 1996) grew a division of labor, trade networks, price, and proto-money from nothing but sign display and reading among coevolving genetic-algorithm agents. SISTER (Duong and Grefenstette 2005) showed that expectations attached to signs carry more meaning than individual recognition, and that knowledge held in mutual expectations survives the death of every member. The present platform is the same engine at modern scale: evolution-strategy and language-model learners for bit-string genomes, a real tool catalog for abstract goods, the market as the institutional field; Section 10 asks the 1991 questions again with modern minds. Multi-agent language-model systems typically assign roles and centralize control; here no role is assigned, and wholesalers are priced into existence rather than prompted.

14 Conclusion

Social algorithms copy the process by which societies construct meaning, as genetic algorithms copy evolution and neural networks copy the brain, and what a social algorithm learns is a society: professions with competing practitioners, supply chains with recorded employment, signs that carry what their bearers deliver, prices learned rather than assigned, and, in a further experiment, whether the society itself is transmissible through its statistics alone.

The reportable result is that decentralized recognition recovers what blind assignment cannot. Market societies outproduced a single long-context mind on identical demands and budgets (0.460 to 0.758 against 0.377, serving up to eight of nine wants against four), and where an allocator's blind assignment recovered nothing of its workers' hidden capability, the market's reward ordering recovered it, a gap of 4.9 ranks against 0.12. The institution, not the group, is what a society adds: prices and signs assign each member to the corner of the field where it earns, with no one knowing the whole. Where capable agents rise regardless of what their unchangeable signs say, the result is an informational defense of open societies: of decentralization, and therefore of democracy, the organization that lets hidden talent become common knowledge.

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ODD Model Description:
Institutions from Interaction

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1 Overview

1.1 Purpose and patterns

The model tests whether institutions such as a division of labor, meaningful signs, prices, supply chains, and employment emerge from symbolic interaction among autonomous learning agents under a designed market mechanism, and whether a society's structure is transmissible to a daughter society through its measured statistics (coevolutionary reconstruction). It is an empirical instrument of the symbolic interactionist paradigm.

Patterns. The model is judged by whether it reproduces, without assignment, the patterns that characterize real market societies: (i) a division of labor with professions practiced by competing specialists (Smith); (ii) supply chains and employment arising from component purchase (Ricardo's comparative advantage); (iii) prices that aggregate information no single agent holds (Hayek); (iv) signaling equilibria, signals that come to carry real information about unobservable quality, as credentials and status displays do in labor markets (Spence); (v) signs acquiring shared meaning, including status differentiation in the mixed-capability and staged-sign experiments, with no dictionary anywhere (symbolic interactionism; the 1991 SDSS lineage); and (vi) transmissibility of social structure through measured statistics alone (the data absorption technique). These patterns double as the evaluation criteria for every result section of the companion paper.

1.2 Entities, state variables, and scales

Learners, evolutionary substrate (24 per standard society; the solitary controls run one): agents whose work is making and selling software, machine-learning programs assembled from a catalog of real tools, and buying components of it from one another. Each carries a genome holding the one sign it displays, a vector of eight floating-point numbers read as a position in meaning space (its public face), and its trade plans: each plan a type (sell, construct, or buy), an item address in the catalog, an offer band, a sought sign of the same eight-number form (the position in meaning space that plan searches), and up to five test slots, each naming a measuring instrument and the corpus to run it on. The test slots are the buyer's side of every trade: whatever a seller delivers to a buy is executed and scored by the buy's tests before any sale can settle, so a seller is paid on measured results, never on its own claims. A buy seeks products; a sell seeks customers. Matching is therefore mutual cosine similarity: the buyer's sought sign against the seller's displayed sign, and the seller's sought against the buyer's displayed, blended half and half. An agent displays exactly one sign. The genome also encodes how deep each plan walks into a tool's parameter settings; a walk of the catalog tree, defined under Details, decodes it all into concrete programs and offers. A per-agent covariance matrix adaptation evolution strategy adapts the genome; genomes never pass between learners, each agent's evolution its own, consistent with the autonomy of perception. Wealth completes the state.

Learners, language-model substrate (11–12 per society): the designed market mechanism and its machinery carry over from the evolutionary substrate unchanged: the same wants, the same catalog item names, the same sell and buy trades, the same double auction, instruments, and payments. What changes is only the mind inside. No genome; a language model writes each period, from a private context, what the genome would have decoded. A sell here is an item name plus the actual code implementing it and an asking price; a buy purchases another learner's component for use in one's own product; the displayed and sought signs are eight-float positions in the same meaning space, written in text. The private context holds the learner's money balance, its recent trade events, a memory string it writes to itself between periods (a private notebook), and the public want sheet: the posted demands, printed with the best score each want has received so far. Both substrates share the same posted demands on the same blackboard; the printed best-score column exists only on the language-model side, where a reading agent needs in text the situation a repeated trader would have experienced, while the evolutionary learner senses the same demand through its own trade outcomes alone. Nothing textual is shared between market learners: consistent with the autonomy of perception, a learner's words reach others only through its displayed sign and its offers. In this study the two substrates do not mix within a society; every market rule applies identically to both.

The coordinator-team control (language-model substrate): the hierarchy the market form is compared against. It has the same eleven workers, wants, instruments, and held-out scoring; a coordinator that assigns work each round from a freshly shuffled anonymous roster showing only each worker's displayed sign and latest delivery result; and a shared scratchpad. Communication by command is the institution under test. In the staged-sign arms it also pays scrip wages that fund bought sign dimensions. It lacks the market: no prices, no offers, no buying or selling between workers, no individual income; the team's output is judged collectively.

Buyers (42 per society, including the solitary controls): the demand field, standing in for the humans the society serves. Each holds one want (item type, test instruments, corpus), an authored budget expressing the degree of want, quality-hunger $\alpha \in [0.3, 0.9]$, a Cobb-Douglas fitness over improvement received and budget kept, and a genome-learned offer band. Buyers are implemented as agents of the same learner class whose want fields (item, instruments, corpus, budget, α) are authored and held fixed at every iteration, while the sign and offer band evolve: the demand is authored, and its prices are learned.

The blackboard: all displayed signs and offers, the only channel between agents. **The catalog**: real machine-learning tools with per-parameter settings levels, 191 production leaves (Section 3.3).

Scales. One iteration is one complete market round (every agent posts, buyers audit, settlements clear, fitness accrues). Standard evolutionary societies hold 66 agents (42 buyers plus 24 learners; the solitary controls hold 43) and run for hundreds to tens of thousands of iterations with no scheduled end; language-model societies hold 11–12 agents and run 20–40 periods. There is no spatial dimension: position exists only in meaning space.

1.3 Process overview and scheduling

Each iteration: agents post messages (signs, sells, buys) from their current plans; buyers gather offers by sign similarity and audit them in sign-similarity-weighted random order, where an audit is the buyer's test on the seller: the offered chain is executed and scored by the buy's instruments on its corpora, a crash or zero score failing, and auditing stops at the first passing delivery (on the language-model substrate, at five audits per want per period); passing deliveries settle under the band double auction (each side's learned offer band; price at the intersection midpoint; defined in full under Details) with quality-scaled payment and ledger entry; fitness signals accrue (sellers: income; buyers: Cobb-Douglas utility over improvement and kept budget; learners buying components for their own products: aspiration-relative satisfaction); each learner advances its own evolution strategy on its own schedule; the board is logged. Within an iteration the partner-choosing step runs twenty passes, so a multi-level supply chain can settle in the round it is offered; agent order is shuffled at every stage.

2 Design concepts

Basic principles. Institutional emergence from symbolic interaction; autonomy of perception (agents are closed with respect to meaning; signs are the only channel; no meaning is copied); double induction (displayers learn what to show from what trades, readers learn what to heed from what follows). **Emergence.** Division of labor, sign meaning (measured as mutual information and geometric separation), price structure, supply chains, employment, and social states are all emergent; none are assigned. A social state is a society's whole standing configuration (which trades are practiced, by how many practitioners, feeding which supply chains, at what prices); the concentrated and dispersed reconstruction sources (Section 3.7) are two such states, distinct enough that a daughter society can be told which one it resembles. **Adaptation and learning.** Each agent adapts through its own learner (CMA-ES or language model); there is no population-level or copied learning. **Objectives.** Sellers maximize income; buyers maximize $\Delta^{\alpha} m^{1-\alpha}$ (improvement over their own adapting aspiration, against budget kept); learner-buyers maximize aspiration-relative satisfaction. **Prediction.** No agent carries an explicit model of future states; language-model agents may deliberate in text, but nothing in the design supplies or rewards forecasting. **Sensing.** Signs and one's own trade outcomes only. Individual recognition, any identity-keyed memory, is banned. Capabilities and internal states are never visible. **Interaction.** Who meets whom is mediated entirely by sign proximity in meaning space on one global blackboard; there is no imposed network topology. Direct interactions are auditions (a buyer examining an offer), sales, and learner-to-learner component purchases; indirect interaction flows through standing offers, prices, and the shared meaning space. Deliveries are tested by real instruments on held-out data, so interaction outcomes are grounded, not declared. **Stochasticity.** Context-keyed reproducible draws: each decision's randomness derives from its context, making replay and resumption exact. **Collectives.** Roles and professions form as sign-organized collectives; societies can seed daughters (reconstruction). **Implementation.** The evolutionary substrate is implemented in Mesa (Python 3.12): the society is a Mesa model, each agent a Mesa agent, learners are CMA-ES via the `cma` package, and the catalog's tools are the real implementations in scikit-learn with gensim and keras variants. The language-model substrate is a standalone Python market loop, not Mesa: the same mechanism stepped directly, each agent's decisions produced by a language model through a provider interface with a complete transcript cache. Exact package versions are pinned in the reproduction kit. **Observation.** Per-iteration boards; reproduction reports; payment ledgers; metrics: division censuses and junk decomposition taken at convergence (each society's last hundred settled deliveries), evenness, sign mutual information and geometry over recent boards, fill order, end-window employment shares, learned-price trajectories, reconstruction fidelities and the diagnostic gap.

3 Details

3.1 Initialization

Genomes uniform random; signs meaningless at birth; budgets set by the authored degree-of-want ladder (in the released configuration files); capability distributions identical across any stuck-sign groups in the mixed-capability experiments.

3.2 Input data

Twelve real text corpora drawn from eight base collections (Internet Research Agency tweets, Internet Research Agency Facebook posts and an IRA-versus-control mix, Trump tweets and a short political-tweets variant, political advertisements, a machine-text detection corpus, congressional bills, disaster tweets, and movie-review sentiment with a few-shot variant, plus a short general variant), and five synthetic vector benchmarks (blobs, moons, circles, anisotropic, varied). Test labels are held back from all graded code. The planted deduplication instrument writes twenty disguised twin rows per targeted corpus (inflections, injected stopwords, doubled letters, pad tokens; deterministic) and scores mutual-nearest recovery against a hidden key. The 42-want demand field references eleven of the seventeen corpus loaders.

3.3 The catalog and the registry tree

The catalog is a tree. Its root has eight branches: two service branches, `data` (17 corpus loaders: the twelve text corpora and five vector benchmarks above) and `test` (8 measuring instruments), and six production branches: `preprocessor` (4 leaves), `clusterer` (49), `vectorSpace` (78), `classifier` (55), `anomalyDetector` (2), and `labeler` (3), for 191 production leaves and 216 leaves in all. An address in the tree is written by underscore-joined levels (`clusterer_sklearn_KMeans_nclusters8`): family, then library, then function, then parameter levels. Parameter levels are themselves tree levels: for each parameterized function the registry declares a small grid of named values, with grids harmonized so the same parameter type walks the same grid across functions (clusterers walk `n_clusters` through 2, 8, 20, and 50, KMeans also `n_init` through 3 and 10; decompositions walk `n_components` through 20, 100, and 300; linear classifiers walk regularization `C` through 0.1, 1.0, and 10.0; random forests walk `n_estimators` through 100, 200, and 500; multilayer perceptrons walk two hidden-layer shapes). A walk that stops before entering a function's parameter levels expresses that function at library defaults, so partial expertise is a first-class address. Tools are instantiated lazily: an address becomes a fitted estimator only when a chain using it is actually evaluated, so the catalog's size costs nothing until the market explores it. Instantiation is restricted to a whitelist of 147 vetted library classes with pinned random seeds; chains are type-checked (a clusterer consumes the numeric output of an embedding stage, not raw text), and inputs are capped at five thousand rows and two thousand columns. One early variant society (always-on seed 210) runs a slightly earlier tree of 182 leaves; every other society shares the 216-leaf tree.

3.4 The genome and its expression

Every learner's genome is a fixed-length vector of 1,148 floats in $[0, 1]$, and its expression into behavior is a deterministic walk. The genome holds a displayed sign (`sign_size` $= 8$ floats in $[0, 1]$, read raw) followed by `num_trade_plans` = 10 trade plans of 114 floats each. Each plan is one type float, its own 8-float sought sign, an item address (`item_size` = 8 floats), midpoint and range floats for the offer band, and `num_tests` = 5 test slots, each two addresses (a measuring instrument from the test branch and a corpus from the data branch) and three flags, one of which ends the test list early, so a plan may carry fewer than five active tests. For the 42 authored buyers the want's tests are fixed; for learners buying components, the test slots are genome-encoded and evolve like everything else. A float expresses a choice by the weighted-level rule: at each tree level the children occupy the unit interval in proportion to their conditional weights (`_weight`), after a stop share (`chance_of_stop_codon` = 0.1) is reserved; the float selects the child whose span contains it, and the walk descends with the next float, stopping when a float lands in stop space or a leaf is reached. The same rule decodes trade types (buy, construct, sell), so the genome is a genotype and the trade plans are its phenotype, exactly in the gene-expression sense. Offer floats map linearly to token prices between 1 and 100. Each learner runs its own covariance matrix adaptation evolution strategy (initial $\sigma = 0.4$, non-elitist, bounds $[0, 1]$, population size at the library default, about 25 at this dimension) over its genome; fitness is market outcome, payments received weighted 1 per token, buyer satisfaction 150, and seller score 100, so selection pressure is the market and nothing else. The population of agents is fixed for a society's life; each learner's generations are the ask-and-tell cycles of its own strategy, advancing on its own schedule.

3.5 The market step in full

Each iteration: (1) every agent posts its expressed trades to the blackboard: 42 buyers each with one want, an authored budget, and a genome-learned offer band; learners with their sells, constructs, and buys. (2) Buyers gather offers whose item matches the want and whose bands overlap, and audit them in sign-similarity-weighted random order (mutual cosine similarity, blended half and half, with a small floor so weak-sign offers are occasionally examined), stopping at the first passing delivery; individual recognition is banned, and no memory attaches to an identity, only to signs. (3) An audited delivery is actually executed: the chain's estimators are fitted and scored by the want's instruments on its corpora; a crash or zero score earns nothing. (4) Settlement is a band double auction: each side's band is its midpoint plus or minus its range; no intersection, no sale; otherwise the price is the intersection's midpoint, scaled by the delivered score over the buyer's frontier (the best score it has verified for that want), so the frontier earns the full agreed price and weaker work earns its fraction. (5) A sold chain that incorporates components bought from other learners cascades payment down the chain (recursion bounded at depth ten), and every transfer lands in the ledger (time, payer, payee, amount). (6) Buyer fitness is the Cobb-Douglas utility $U = \Delta^{\alpha} m^{1-\alpha}$, where Δ is the satisfaction delta (score above the buyer's aspiration, an exponential moving average of its accepted quality with rate 0.05, floored at zero) and m is the share of budget kept; α is spread 0.3 to 0.9 across buyers, and budgets run 12 to 130. Prices are therefore learned at both ends and nothing is scheduled: token prices live between 1 and 100, budgets express the authored degree of want, and every posted want stands from birth.

3.6 The language-model substrate in full

The same market mechanism with the learner replaced: each agent is a language model that reads its situation each period (its balance, its recent events, its private notebook of up to 1,200 characters, a board view of its four sign-nearest sellers, and the public want sheet with best scores) and answers with trades and code. Agents start with 150 tokens of wealth and pay small metered costs for their own context and output, so deliberation itself has a price. Deliveries are graded in a sandbox by the same held-out instruments; labels live in a truth directory the graded code cannot read, and predictions are scored harness-side. Full-mechanism societies, called the school societies below, run twelve agents for forty periods, with the aspiration bar (a buyer passes over deliveries below its aspiration, an exponential moving average of accepted quality) and component payments as above. The mixed-capability arms run eleven agents drawn from a hidden cast of five small, four middle, and two large models (Haiku, Sonnet, and Opus tiers), shuffled per society, identical across paired forms, with capability visible nowhere: not to agents, not to any coordinator. The market form runs the standard market; the team form replaces it with a coordinator that sees only a freshly shuffled anonymous roster of (sign, last delivery) each round and assigns work to positions, so signs are the only stable public face anywhere. Worker signs are eight numbers in $[-1, 1]$, self-chosen and freely changeable each round in arm A; in arm B two dimensions display only at a cost of 2 per unit intensity per round, paid from market wealth or, in the team form, from a scrip wage of 10 per scored delivery, and zeroed if unaffordable; in arm C one further dimension is fixed at birth to $+1$ or -1, stratified so capability tiers split evenly between markers. The staged-sign arms read at a declared twenty-period checkpoint. Per-round sign records are logged against the hidden cast for the sorting analyses. Every model call is cached to disk by prompt, so a completed society replays deterministically; the mixed-capability program cost about \$376 at list prices.

3.7 Measurement instruments, each defined by its computation

Division census. A learner's trade is its most frequent settled delivery, attributed by the tool parts of paid chains; societies are censused as distinct trades, practitioners per trade, and evenness (normalized entropy over practitioners per trade). Where a society's report spans restarts, archived and current segments are concatenated in chronological order and replayed rows are deduplicated before the census.

Employment shares. From the payment ledger (time, payer, payee, amount): agent roles are read from the latest board (an agent with a utility-bearing buy is a buyer; a seller otherwise is a learner); the end window is the final quarter of ledger time; the statistic is the learner-to-learner share of payment count and of payment value, end-window and lifetime, on ledgers with at least forty payments.

Settle order. Each want's tier comes from its configuration: entry (one instrument), multi-instrument (two), hard (three or more, or two corpora). The statistic is each want's first settlement iteration, summarized as the per-tier median, and the ordering test asks whether entry precedes multi precedes hard.

Junk decomposition. A junk settlement scores at or below 0.05. Reported per tier (entry versus upper) as the junk share of settlements.

Sign informativeness (displayed signs only, by the author's ruling). Over each society's last thirty boards, unique (seller, item) sell listings are collected, each carrying the seller's displayed sign: the one public sign at the top of its message, never a plan's sought sign. Displayed signs are quantized by 8-means clustering within the window; the statistic is the mutual information in bits between the sign cluster and the listed item's family. The design expectation tested is Spence's: signals stay informative while maintaining them is costly to the unqualified, and collapse when payment stops tracking quality.

Finest-grain sign similarity (displayed signs, different sellers). Same listings; pairs of same-role listings from different sellers are labeled within-family (same tool, different settings) or between-family (different tools); the statistic is the mean cosine similarity difference of the sellers' displayed signs (within minus between), permutation over family labels within role. Same-seller pairs are excluded because an agent displays exactly one sign.

Task-family sign geometry. Per-seller displayed sign (its latest within the window), sellers grouped by primary family (representation: embedding and clustering; anomaly detection); the statistic is mean within-representation cosine minus representation-to-anomaly cosine, permutation over role labels.

Learned-price tiers. Buyer offer midpoints averaged by tier (entry versus composite) on the first and the latest board; the registered prediction is a widening composite premium.

Software quality per society. Per want, the running best score from the report (unfilled wants count zero); a society's quality is the mean best-per-want, comparable at a common age (equal market rounds) or at common total learner-effort (rounds times learners). The solitary-control comparison uses both framings.

Wholesaler criterion (declared before the unseen pass). A society has a wholesale supplier if some agent reaches at least 30 component sales, at least a 30 percent share of the society's component-market sales, and at least 5 distinct consumers. The four school societies calibrate the criterion; the mixed-capability market societies are the unseen test set, with the declared moderator that the test societies run twenty periods and eleven agents rather than forty and twelve.

Reconstruction panel (registered). Sources chosen by maximal total-variation distance between full practice profiles (each society's distribution of practiced trades) among the eldest societies. Census: every tool appearing in paid deliveries of the source's final five iterations, counted over the production branches only. Seeding: at every catalog tree level, branch conditional probabilities arranged so ten percent of the tool space stays uniform and ninety percent follows the census; a source's use of a tool at library defaults credits the tool choice, never its settings variants; data and instrument branches keep natural uniform weights. References frozen at seeding. Read at daughter age 300 or more: each daughter's own standing-state census against both frozen sources; similarity is total-variation similarity; fidelity is similarity to own source minus similarity to the other; unseeded controls define background drift; the diagnostic gap (seeded versus sustained frequencies) is registered to report alongside.

3.8 Outputs written by the model

Every settlement writes the board; every transfer writes the payment ledger (time, payer, payee, amount); every generation writes a reproduction report (time, agent, fitness, tokens, scores, displayed sign, bought items). Language-model societies log every period to a run journal. All claims in the companion paper trace to these files.

3.9 Model code and data availability

The engine, every society configuration, the language-model market harness, every analysis instrument, the hidden capability casts, and the frozen reconstruction references are public at `https://github.com/agentbasedlearningsystems/institutions-from-interaction` (Agent Based Learning Systems, `https://agentbasedlearningsystems.com`). The complete per-society results tables and the experiment registration ledger behind the companion paper are distributed with the repository; full run outputs (reports, boards, payment ledgers, transcripts; tens of gigabytes) are retained in full and available on request.